

MASTER

Master in Applied Econometrics and Forecasting

MASTER'S FINAL WORK

DISSERTATION

*Regime-Switching Dynamics in E-commerce Inventory:
Evidence from Amazon FBA Daily Ledger Data*

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ABSTRACT, KEYWORDS AND JEL CODES

This thesis investigates whether nonlinear regime-switching time series models improve upon standard linear benchmarks for daily stock-level forecasting in an Amazon Fulfilment by Amazon (FBA) setting. The empirical analysis uses Amazon SP-API inventory ledger data provided by ScalerPath, covering eight anonymised Master SKUs with approximately 108 daily observations each. Five competing models are estimated and compared: a naïve benchmark, ARIMA, exponential smoothing (ETS), a Self-Exciting Threshold Autoregressive (SETAR) model, and a Markov-Switching Autoregressive (MS-AR) model. Forecast accuracy is evaluated using a rolling-origin expanding-window design at horizons of one, seven, and fourteen days, with performance measured by RMSE and MASE. Statistical comparison is conducted using Diebold-Mariano tests and the Model Confidence Set procedure. Results show that no single model dominates across all SKUs and horizons. Linear models remain competitive at short horizons, while MS-AR demonstrates stronger relative performance at longer horizons for several SKUs. SETAR produces unstable multi-step forecasts in a subset of cases, requiring a relative instability filter. MS-AR is never eliminated from the superior set in the valid Model Confidence Set results. In addition, a parametric Monte Carlo simulation from the fitted MS-AR model is used to estimate the probability of entering a relative low-stock state, revealing heterogeneous risk profiles across SKUs that are not captured by point forecasts alone. This risk measure is interpreted as a relative low-stock indicator rather than a physical stockout probability. The study contributes an empirical application of regime-switching forecasting models to daily FBA inventory stock data and shows that MS-AR models can provide both competitive point forecast accuracy and interpretable probabilistic low-stock risk assessments relevant to replenishment decision-making.

KEYWORDS: inventory forecasting, regime-switching, Markov-switching autoregressive model, SETAR, Amazon FBA, rolling-origin evaluation, Model Confidence Set, low-stock risk

JEL CODES: C22; C53; M11

Declaration on the Use of Artificial Intelligence

Artificial intelligence tools were used in the conduct of this study to facilitate the development and debugging of the R code. The methodological approach, empirical analysis, interpretation, and final content of the following dissertation are the responsibility of the author.

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1 INTRODUCTION

The expansion of e-commerce over the past decade has transformed inventory management from a back-office operational function into a strategic capability. As sellers increasingly rely on platform fulfilment infrastructure, in particular Amazon’s Fulfilment by Amazon (FBA) programme, the ability to forecast stock levels accurately and anticipate low-stock episodes has become central to maintaining service levels, preserving marketplace performance, and controlling working capital costs. Under FBA, inventory is stored within Amazon’s warehouse network and replenishment orders must be planned in advance. Forecasting errors can therefore translate directly into either stockout episodes, which reduce sales and may negatively affect marketplace performance, or excess stock, which incurs storage costs and ties up capital. Despite the operational importance of this problem, most time-series forecasting research applied to e-commerce focuses on sales demand rather than on-hand inventory. This distinction matters. Observed sales may be censored by inventory availability: when inventory is unavailable, sales no longer reveal latent demand. Sales also do not directly indicate whether the inventory position is approaching a low-stock state. Modelling stock levels directly provides a more natural representation of the replenishment decision problem and is made feasible by the granular daily inventory ledger data available through the Amazon SP-API.

A further feature of Amazon FBA inventory series is that they do not evolve smoothly over time. Visual inspection of daily stock data reveals two qualitatively different dynamic patterns: extended periods of gradual, persistent depletion and short-lived episodes of sharp adjustment. The available data do not identify the operational events generating these adjustments, which may reflect replenishment activities, inventory transfers, inventory reconciliations or other warehouse operations. This behaviour suggests that nonlinear time-series models may provide additional insight into the inventory dynamics. Two such model classes are considered in this study: the Self-Exciting Threshold Autoregressive (SETAR) model, which identifies regimes through an observed stock threshold, and the Markov-Switching Autoregressive (MS-AR) model, which treats regimes as latent states governed by transition probabilities.

This thesis investigates whether nonlinear regime-switching specifications improve upon standard linear benchmarks for daily SKU-level inventory forecasting. The empirical analysis uses Amazon SP-API inventory ledger data provided by ScalerPath, an operational partner specialising in Amazon e-commerce management, covering eight anonymised Master SKUs

with complete daily observations over a sample of approximately 108 days. Five competing models are estimated and compared: a naïve benchmark, ARIMA, exponential smoothing (ETS), SETAR and MS-AR. Forecast accuracy is evaluated using a rolling-origin expanding-window design at horizons of one, seven and fourteen days, with performance measured by RMSE and MASE. Statistical comparison is conducted using Diebold-Mariano tests and the Model Confidence Set procedure. In addition, the fitted MS-AR model is used to simulate future inventory paths and estimate the probability that inventory is below a relative low-stock threshold at one-, seven-, and fourteen-day forecast horizons. These probabilities provide a relative low-stock risk indicator that complements traditional point forecasts.

The study makes three contributions. First, it provides an empirical evaluation of regime-switching forecasting models on daily FBA inventory stock data, a setting that has received limited attention in the academic literature. Second, it combines standard forecast accuracy evaluation with a Monte Carlo simulation framework for relative low-stock risk assessment, extending point forecasts with horizon-specific probabilistic risk estimates. Third, it examines whether MS-AR models can provide interpretable regime characterisations of inventory dynamics, distinguishing stable, persistent periods from volatile adjustment episodes and complementing quantitative forecasts with operational insight.

The remainder of the thesis is structured as follows. Section 2 reviews the relevant literature on inventory forecasting, linear and nonlinear time-series models, and forecast evaluation methods. Section 3 describes the data source, preprocessing steps and descriptive statistics. Section 4 presents the methodology, covering model specifications, the rolling-origin evaluation design, accuracy metrics, statistical comparison procedures and the low-stock risk simulation. Section 5 reports the empirical results. Section 6 discusses the findings, limitations and directions for future research, and concludes the thesis.

2 LITERATURE REVIEW

2.1 Forecasting in Inventory Management

Accurate inventory forecasting is a central problem in operations management and supply chain planning. The ability to predict future stock levels - or the demand that drives them - determines replenishment decisions, safety stock policies, and service level outcomes. The operations management literature has long recognised that forecasting errors propagate into inventory costs: understocking leads to lost sales and stockouts, while overstocking generates holding costs and capital tie-up (Silver, Pyke, and Thomas 2017). Classical inventory models, such as economic order quantity and reorder-point frameworks, often rely on assumptions about the probability distribution of future demand, lead times, and service-level targets. In practice, however, inventory series often exhibit serial dependence, persistence, and occasional abrupt adjustments. These changes may reflect replenishment activities, promotional events, supply disruptions, inventory reconciliations, or other operational processes. These features make time-series forecasting methods a natural complement to classical inventory theory. Syntetos, Boylan and Disney (2009) provide a comprehensive review of forecasting methods for inventory planning, documenting sustained interest in univariate statistical approaches alongside more recent machine learning extensions. The growth of e-commerce and fulfilment-by-marketplace programmes such as Amazon FBA has intensified the operational importance of inventory forecasting. In these settings, sellers must anticipate stock needs sufficiently in advance to avoid stockouts within the platform's warehouse network. The daily inventory ledger data generated by such systems provide a rich time series of observed inventory levels suitable for statistical modelling and out-of-sample forecast evaluation.

2.2 Linear Forecasting Models

2.2.1 ARIMA

The autoregressive integrated moving average (ARIMA) model, developed systematically by Box and Jenkins and described in Box et al. (1994), remains one of the most widely used frameworks for univariate time-series forecasting. ARIMA models capture serial dependence

through autoregressive terms, short-run error dynamics through moving-average terms, and non-stationarity through differencing. In compact notation, an ARIMA(p, d, q) process satisfies

$$\Phi(L) (1-L)^d X_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2),$$

where $\Phi(L)$ and $\Theta(L)$ are lag polynomials of order p and q , and $(1 - L)^d$ denotes the differencing operator. Stationarity requires all roots of $\Phi(z) = 0$ to lie outside the unit circle; invertibility requires the same for $\Theta(z) = 0$ (Chatfield and Xing 2019, chap. 3). The principal limitation of ARIMA is that the same linear dynamic structure is assumed to hold throughout the sample, which may be restrictive when inventory series exhibit abrupt changes in their dynamics. Automatic selection procedures based on information criteria, such as that of Hyndman and Khandakar (2008), have made ARIMA a practical and widely adopted benchmark in comparative forecast studies.

2.2.2 Exponential Smoothing and ETS Models

Exponential smoothing methods generate forecasts as geometrically weighted averages of past observations. Simple exponential smoothing (SES) is closely related to an ARIMA(0,1,1) model without drift, which helps explain its empirical usefulness across a wide range of persistent series without clear trend or seasonality (Chatfield and Xing 2019, sec. 5.2). Extensions to Holt’s two-parameter method and the Holt-Winters three-parameter procedure accommodate trends and seasonal components. The modern ETS (Error-Trend-Seasonality) framework, reviewed in Hyndman and Athanasopoulos (2021), unifies all exponential smoothing variants within a state-space model, distinguishing error type (additive or multiplicative), trend (none, additive, or damped), and seasonality (none, additive, or multiplicative).

Model selection across variants using information criteria makes ETS a competitive and principled forecasting benchmark. In inventory contexts, ETS is particularly relevant because stock levels often evolve smoothly between adjustment events, which the level-updating equations capture naturally.

2.3 Nonlinear Time Series Models

Linear ARIMA and ETS frameworks assume a constant dynamic structure throughout the sample. This assumption may be violated when a time series alternates between qualitatively different regimes - for example, periods of stable depletion and periods of abrupt adjustment. Nonlinear time series models relax this assumption by allowing the parameters of the process to depend on the state of the system (Chatfield and Xing 2019, chap. 11).

2.3.1 Threshold Autoregressive Models

Threshold autoregressive (TAR) models, introduced and developed extensively by Tong (1990), specify that the autoregressive parameters switch according to whether an observable state-determining variable crosses a threshold. When the threshold variable is a lagged value of the series itself, the model is called self-exciting, yielding the SETAR model. A two-regime SETAR with delay d and threshold γ takes the form:

$$X_t = \begin{cases} \alpha_1 + \phi_{1,1}X_{t-1} + \dots + \phi_{1,p}X_{t-p} + \varepsilon_{1,t}, & X_{t-d} < \gamma \\ \alpha_2 + \phi_{2,1}X_{t-1} + \dots + \phi_{2,p}X_{t-p} + \varepsilon_{2,t}, & X_{t-d} \geq \gamma \end{cases}$$

Threshold models provide a piecewise linear approximation to general nonlinear dynamics, with regime membership determined by an observable variable (Tong 1990; Chatfield and Xing 2019, sec. 11.3). Chappell et al. (1996) demonstrated improved out-of-sample forecast accuracy from a SETAR specification over linear and random walk benchmarks in an exchange rate application, providing empirical motivation for threshold models in short-horizon forecasting.

2.3.2 Markov-Switching Autoregressive Models

Markov-switching autoregressive (MS-AR) models, introduced by Hamilton (1989), extend the standard AR model to allow regimes governed by a latent Markov chain rather than an observable threshold. Let $s_t \in \{1, \dots, K\}$ be an unobserved regime variable following a first-order Markov chain with transition matrix $P = (p_{ij})$, where $p_{ij} = P(s_t = j \mid s_{t-1} = i)$. Conditional on $s_t = j$, the process satisfies:

$$X_t = \alpha_{s_t} + \phi_{s_t}X_{t-1} + \varepsilon_{s_t,t}, \quad \varepsilon_{s_t,t} \sim N(0, \sigma_{s_t}^2)$$

where the intercept, autoregressive coefficient, and innovation variance are all regime-specific. The latent regime structure is particularly appealing for inventory stock series, where abrupt changes may reflect replenishment activities, warehouse transfers, inventory reconciliations, or other operational processes that are not directly observable in the stock series. Parameters are estimated by maximum likelihood, using filtering and smoothing recursions to infer the latent regime probabilities (Kim and Nelson 1999). The smoothed regime probabilities $P(s_t = j \mid X_1, \dots, X_T)$ provide a probabilistic characterisation of inventory states over the entire sample, enabling post-estimation regime interpretation. Hamilton (1989) applied MS-AR models to US real GNP and identified two regimes corresponding to expansions and contractions. Engel and Hamilton (1990) extended the framework to exchange rates, while Bergman and Hansson (2005) demonstrated that two-regime MS-AR models can produce lower

mean squared forecast errors than random walk benchmarks at horizons up to four steps ahead, a finding that motivates the application of MS-AR to inventory data in the present study. More recently, the relevance of regime-switching dynamics for inventory and production management has been explored in the operations research literature. Cadenillas, Ferrari and Schuhmann (2024) study optimal production control under a continuous-time Markov chain that modulates demand parameters, showing that regime-dependent modelling can be relevant for production and inventory control decisions.

2.4 Forecast Evaluation and Statistical Comparison

2.4.1 Accuracy Metrics

The assessment of forecast accuracy requires metrics that are interpretable and comparable across models and series. Root mean squared error (RMSE) penalises large errors more heavily due to the squaring of residuals and is expressed in the same units as the original series. However, RMSE is scale-dependent and cannot be used to compare performance across series with different inventory levels. Hyndman and Koehler (2006) propose the mean absolute scaled error (MASE) as a scale-free alternative. MASE expresses the model's mean absolute error relative to the mean absolute in-sample naïve error, yielding an interpretable benchmark: a MASE below one indicates improvement over the naïve. MASE is wellbehaved even for persistent and near-zero series and has become standard in academic forecast evaluation (Hyndman and Athanasopoulos 2021).

2.4.2 Diebold-Mariano Test

Comparing average accuracy metrics does not establish whether performance differences are statistically significant. Diebold and Mariano (1995) propose a test of the null hypothesis of equal predictive accuracy between two competing models. The test is based on the loss differential $d_t = L(e_{1t}) - L(e_{2t})$, where $L(\cdot)$ is a loss function, and e_{jt} is the forecast error of model j . Under the null, $E[d_t] = 0$, and the DM statistic is asymptotically standard normal under general conditions on the loss differential process. For multi-step forecasts, the loss differential may be serially correlated because overlapping rolling forecast errors share common information. Consistent estimation of the long-run variance of the loss differential is therefore required for valid inference. West (1996) provides a general asymptotic framework for inference about predictive ability using out-of-sample prediction errors, allowing for non-nested and nonlinear models as well as serial dependence, and establishes conditions under

which standard normal inference remains valid. HAC estimators, including Bartlett-kernel estimators (Andrews 1991), provide a standard approach to estimating the required long-run variance in multi-step forecast comparison settings.

2.4.3 Model Confidence Set

The Diebold-Mariano test is a pairwise procedure and does not account for the multiple testing problem that arises when comparing several models simultaneously. Hansen, Lunde and Nason (2011) propose the Model Confidence Set (MCS) as a solution. The MCS identifies the set of models $\mathcal{M}^*$ such that, with confidence level $1 - \alpha$, the best model belongs to $\mathcal{M}^*$. The procedure sequentially eliminates models that are significantly inferior to the best remaining model, using bootstrap-based critical values. The MCS has become a standard tool in forecast comparison studies because it provides a principled answer to whether a single model is distinguishable from the best, or whether several models perform equivalently.

2.5 Research Gap and Contribution

The application of forecasting methods to warehouse stock management is a topic that has been extensively studied in the literature, particularly in the context of demand forecasting and replenishment planning. In this context, Syntetos, Boylan and Disney (2009) provide a comprehensive review of the main forecasting methodologies for inventory planning, highlighting how the accuracy of forecasts is a key factor in supporting effective inventory management decisions. The literature also focuses on the analysis of non-linear and regime-dependent dynamics in production and inventory management problems. For example, Cadenillas, Ferrari and Schuhmann (2024) study an optimal production and inventory control problem in which the parameters evolve according to a regime-switching process, demonstrating how the introduction of different regimes can describe the system's evolution more realistically. In the specific context of e-commerce, Hu, Hu and Li (2025) propose a joint framework for forecasting sales and stock levels within the Amazon ecosystem, confirming the growing interest in forecasting models applied to digital platforms.

The reviewed literature provides well-established tools for inventory forecasting, linear time-series modelling, nonlinear regime-switching models, and statistical forecast comparison. However, relatively little attention has been given to the use of regime-switching time-series models for daily SKU-level inventory stock data generated by platform fulfilment systems such as Amazon FBA. Most inventory forecasting applications in the literature focus on demand or

sales, whereas this study models sellable stock levels directly. This distinction is important: observed stock levels reflect not only demand but also replenishment, warehouse transfers, inventory reconciliations, and other operational adjustments. Modelling stock directly provides a natural framework for evaluating low-stock exposure within a replenishment-relevant horizon, without requiring additional assumptions to recover inventory from sales data. Some recent reviews of retail e-commerce forecasting also document a broad shift toward machine learning and deep learning architectures for high-dimensional SKU-level demand settings. However, many of these approaches are based on complex machine learning architectures, whose internal decision-making process is less interpretable than that of traditional statistical time-series models. They are less suited to settings where interpretability is important, sample sizes are modest, or covariate data are unavailable. This study contributes by evaluating whether interpretable statistical regime-switching models, SETAR and MS-AR, can improve upon linear benchmarks for daily SKU-level stock forecasting in a real Amazon FBA operational setting. The analysis evaluates both point forecast accuracy across multiple horizons and probabilistic low-stock risk, thereby linking time-series regime-switching methods with an operational inventory management problem. The literature reviewed shows that regime-switching models have been successfully applied in various economic and financial contexts and, more recently, to problems relating to production and inventory management. However, the sources considered in this paper do not reveal any direct applications to the operational forecasting of stock levels using daily data from Amazon FBA ledgers.

3 DATA

3.1 Data Source

The data used in this study are extracted from the Amazon Seller Partner API (SP-API) inventory ledger report for a seller operating under the Fulfilment by Amazon (FBA) programme. Under FBA, Amazon stores, picks, packs, and ships products on behalf of the seller. The inventory ledger provides daily snapshots of onhand inventory broken down by product, fulfilment centre location, and inventory disposition. Disposition refers to the condition classification of inventory units: categories include SELLABLE, UNSELLABLE, RESERVED, and others. Only SELLABLE units - those available for immediate customer order fulfilment - are included in this analysis. The raw data contain one record per SKU, warehouse location, and date, with the field `Final_OnHand_Effective` recording the end-of-day

stock count for that unit. Since a single product may be distributed across multiple fulfilment centres simultaneously, daily SKU-level stock is obtained by summing `Final_OnHand_Effective` across all warehouse locations for each Master SKU and date. The resulting time series represents the total available SELLABLE stock for each SKU on each day, which is the quantity relevant for assessing aggregate replenishment risk. Although the raw ledger therefore records inventory at the SKU-location-date level, the empirical analysis aggregates stock across fulfilment locations for each Master SKU and date. This choice reflects the objective of forecasting aggregate sellable inventory availability at the product level rather than modelling warehouse-specific logistics. Given the short sample length, modelling each SKU-location pair independently would substantially reduce the number of usable observations per series and increase estimation instability, especially for regime-switching models.

3.2 Sample Period and Observation Count

The empirical analysis was conducted over the period from 22 January 2026 to 10 May 2026. Each of the selected SKUs contains 108 daily observations. The original dataset contains only one missing value relating to 14 April 2026; otherwise, all the remaining selected SKUs share the same observed daily time index. The sample spans roughly 15 calendar weeks. This sample size is adequate for a parsimonious comparison of autoregressive and two-regime forecasting specifications, but it also imposes important constraints. Long seasonal cycles cannot be estimated reliably, and models with many parameters must be restricted to reduce the risk of overfitting. These constraints are reflected in the model specifications described in Section 4.

3.3 SKU Selection

The raw dataset contains 36 Master SKUs. However, not all series are suitable for regime-switching time-series modelling, given the short sample length and the need for complete, positive, and sufficiently variable daily stock observations. The full seller catalogue therefore contains more SKUs than can be meaningfully modelled with the available sample length. Regime-switching models in particular require enough observations within each estimated regime to identify regime-specific dynamics. Series that are sparse, near-zero

for extended periods, discontinuous, or almost constant provide limited information for estimating distinct inventory behaviours. Eight Master SKUs were selected according to three criteria. First, each SKU must have complete daily coverage across the entire sample period, with no missing dates. Second, each SKU must exhibit a positive and economically meaningful

average stock level, excluding products that were temporarily out of stock for most of the sample. Third, each series must display sufficient time-series variation to support estimation of autoregressive dynamics. SKUs with near-constant stock throughout the sample were excluded because they provide little information for comparing forecasting models or identifying regime changes. The eight selected SKUs are identified throughout this study by anonymised codes: AA00, AA03, AA04, AG00, AG01, AG03, AG04 and AJ00. Product identities are masked to preserve commercial confidentiality. The selected SKUs represent products with consistently positive stock levels and heterogeneous inventory dynamics, making them appropriate for comparing linear benchmarks against nonlinear regime-switching models.

3.4 Descriptive Statistics

Table I: Descriptive statistics of daily SELLABLE stock levels

SKU	n	Mean	SD	Min	Max	CV
AA00	108	6,090.9	1,032.6	3,961	8,244	0.170
AA03	108	3,595.1	957.1	2,200	5,519	0.266
AA04	108	3,330.3	590.3	1,839	4,225	0.177
AG00	108	4,896.6	737.3	2,858	5,886	0.151
AG01	108	1,426.8	253.6	851	1,923	0.178
AG03	108	2,860.0	526.9	1,661	4,055	0.184
AG04	108	2,683.4	506.9	1,500	3,232	0.189
AJ00	108	2,231.0	341.9	1,646	2,849	0.153

Table 1 reports descriptive statistics for the eight selected SKU series. All SKUs contain 108 daily observations, confirming that the selected sample is balanced and that each series shares the same daily time index. Mean stock levels differ substantially across products, ranging from 1,426.8 units for AG01 to 6,090.9 units for AA00. This confirms that the selected SKUs operate at different inventory scales. The coefficient of variation provides a scale-free measure of variability. The lowest coefficients of variation are observed for AG00 (0.151) and AJ00 (0.153), indicating relatively stable inventory levels compared with their average stock. The highest coefficient of variation is observed for AA03 (0.266), suggesting greater relative variability over the sample period. The remaining SKUs have coefficients of variation between

approximately 0.17 and 0.19, indicating moderate but non-negligible inventory fluctuations. The minimum and maximum values also show that none of the selected SKUs reaches extremely low or zero stock levels during the sample. Across the SKU-specific minimum values, the lowest observed stock is 851 units for AG01, while the highest SKU-level minimum is 3,961 units for AA00. This is important for interpretation: the analysis does not study physical stockouts, but rather relative low-stock states within each SKU’s historical inventory distribution. Overall, the descriptive statistics confirm that the selected SKUs differ in scale and relative variability. These differences motivate the use of SKU-specific model estimation and scale-adjusted accuracy metrics such as MASE in the forecast evaluation.

3.5 Preliminary Data Patterns

Figure 1: Daily SELLABLE stock trajectories for the eight selected Master SKUs

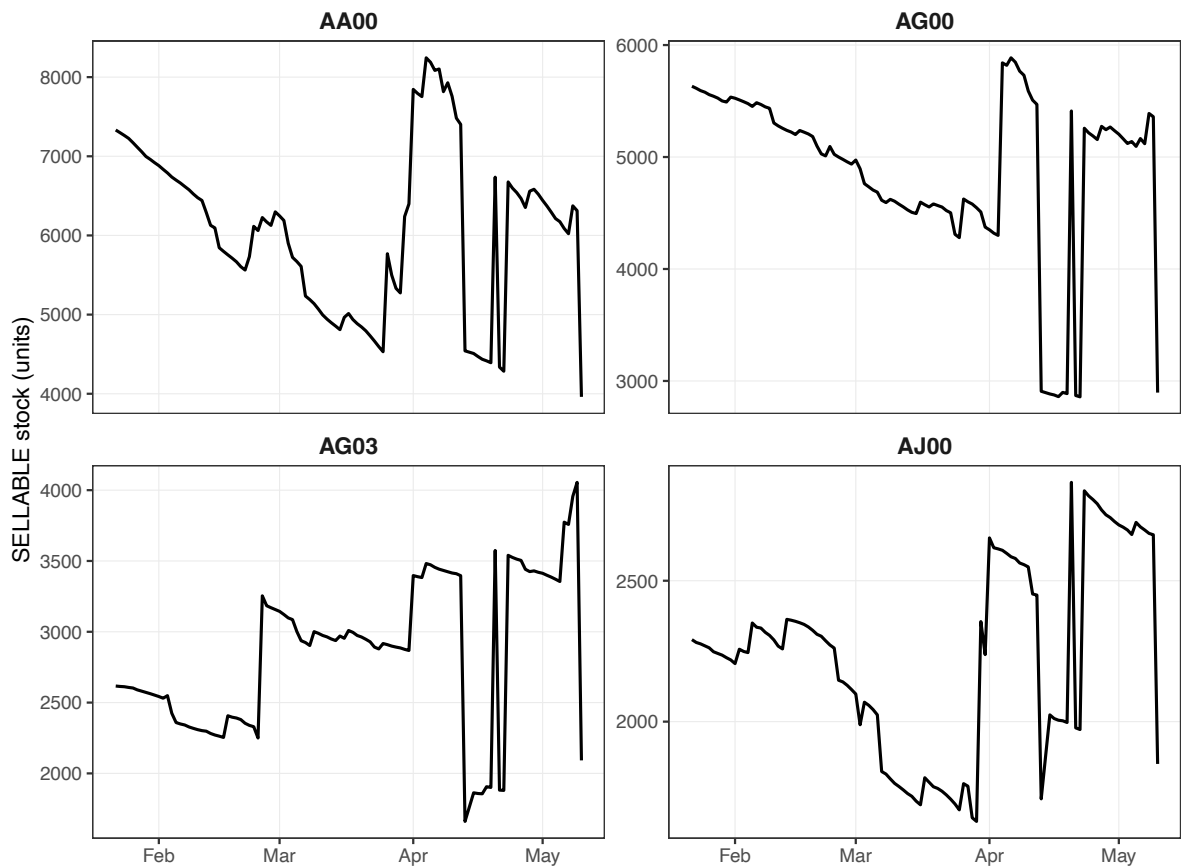


Figure 1 shows the daily SELLABLE inventory trajectories for four representative Master SKUs. Two common features emerge across the selected series. First, inventory levels often evolve gradually over consecutive days, indicating a high degree of persistence. Second, this

gradual behaviour is occasionally interrupted by abrupt upward or downward changes, producing distinct shifts in the observed inventory level. The available data do not identify the operational events generating these changes. They may reflect replenishment activities, inventory transfers, stock adjustments, or other warehouse operations. Regardless of their origin, these abrupt movements suggest that the inventory process does not evolve under a single stable dynamic throughout the sample. These observations suggest that exploring nonlinear time series models may provide additional insight into the inventory dynamics. Among these approaches, the SETAR model allows the dynamics to change when inventory crosses an observed threshold, whereas an MS-AR model assumes that the underlying regime is latent and evolves according to transition probabilities.

FIGURE 2: ACF (left) and PACF (right) for each SKU.

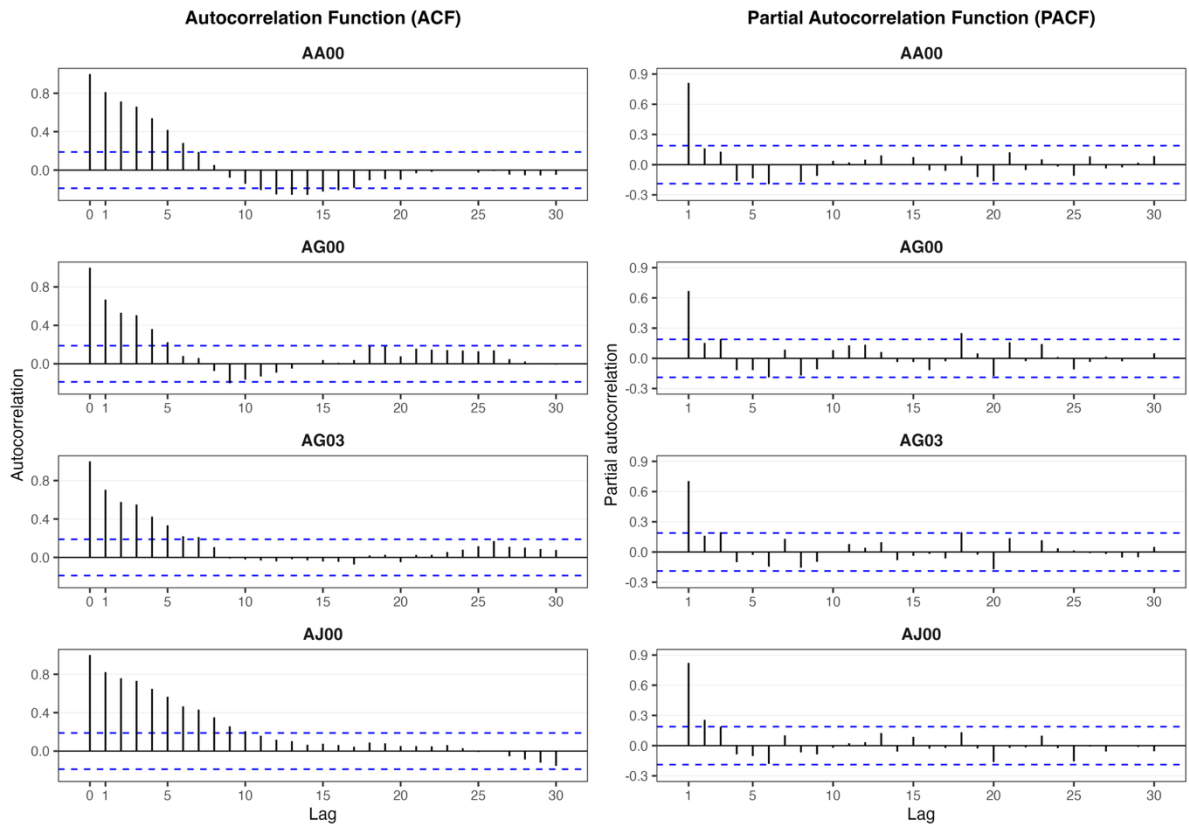


Figure 2 supports these observations using the autocorrelation structure. The ACF displays a slow decay for all selected SKUs, indicating strong persistence in inventory levels. The PACF is dominated by the first lag, while most remaining lags lie within the confidence bands, suggesting that much of the serial dependence is explained by the most recent observation. This pattern is consistent with a low-order autoregressive component and supports the use of autoregressive specifications in the subsequent forecasting models.

4 METHODOLOGY

4.1 Empirical Design

The empirical strategy follows three steps. First, daily inventory stock series are constructed at the SKU level from Amazon FBA inventory ledger data. Second, five competing forecasting models are estimated and compared: a naïve benchmark, ARIMA, ETS, SETAR, and MS-AR. The first three models provide standard linear or smoothing benchmarks, while SETAR and MS-AR are included to assess whether nonlinear regime-dependent dynamics improve forecasting performance. Third, the best-performing regime-switching model is used to simulate future stock paths and estimate relative low-stock risk. This structure reflects the objective of the study. The analysis is not limited to comparing point forecasts; it also evaluates whether regime-switching models provide operationally useful information about inventory states and low-stock exposure. For this reason, the methodology combines standard forecast evaluation, statistical forecast comparison tests, and a probabilistic simulation exercise. The unit of analysis is the aggregate Master SKU time series. This choice is primarily methodological and operational. With only approximately 108 daily observations, modelling each SKU-location pair separately would substantially reduce the number of usable observations per series and increase estimation instability, especially for regime-switching models. Aggregating across fulfilment locations therefore provides a more parsimonious and stable basis for comparing forecasting models. At the same time, the aggregate Master SKU series remains managerially informative. It captures the overall stock level and general inventory dynamics of each product across the FBA network, rather than warehouse-specific allocation decisions. Aggregate SKU forecasts can help identify which products are experiencing persistent depletion, which products are approaching a relative low-stock state, and which SKUs require closer monitoring or replenishment prioritisation. The aggregate approach therefore provides an operationally interpretable view of product-level inventory behaviour, while leaving warehouse-level allocation analysis for future work with longer SKU-location histories. In this analysis, X_t denotes the daily aggregate level of SELLABLE stock for a specific Master SKU on day t , calculated by summing the units available across the various FBA fulfilment centres.

4.2 Why Stock Levels Rather than Sales

The dependent variable in this study is daily on-hand inventory rather than sales volume. This choice is motivated by the operational objective of the analysis. The main question is whether a SKU is likely to enter a low-stock state within a replenishment-relevant horizon. This question is naturally expressed in terms of stock levels, since replenishment decisions depend on the current and expected future inventory position. Sales are more direct measures of demand, but they are not equivalent to inventory risk. Observed sales may be censored by availability: when stock is unavailable, sales can fall to zero even if latent demand remains positive. As a result, a sales-based model would require additional assumptions to distinguish low demand from lost sales caused by stock constraints. By modelling stock directly, the analysis focuses on the inventory state variable that determines low-stock exposure. Stock levels are also expected to display stronger persistence than daily sales, because inventory evolves through gradual depletion interrupted by replenishment or adjustment events. This persistence makes stock levels suitable for autoregressive and regime-switching time-series models.

4.3 Preliminary Time-Series Analysis

4.3.1 Stationarity

The ADF and Phillips-Perron tests use a unit-root null hypothesis, while the KPSS test uses a level-stationarity null hypothesis. PP rejects the unit-root null when the statistic is more negative than the 5 percent critical value. The tests produce mixed evidence across SKUs. Some series show evidence consistent with stationarity in levels, while others appear highly persistent or borderline non-stationary. Rather than applying a uniform differencing transformation, the regime-switching models are estimated in levels. This choice is central to the research objective: SETAR thresholds and MS-AR regimes are intended to describe stock-level states. Differencing would remove the level information needed to interpret lower- and higher-inventory regimes and would make the subsequent low-stock risk simulation less directly interpretable.

TABLE II: Stationarity tests: ADF, KPSS and Phillips-Perron

SKU	ADF p-value	KPSS p-value	PP statistic	PP 5% cv	PP reject
AA00	0.2431	0.1000	-2.80	-2.89	No
AG00	0.0334	0.0300	-3.89	-2.89	Yes
AA03	0.3357	0.0110	-2.57	-2.89	No
AA04	0.2281	0.0932	-3.74	-2.89	Yes
AG03	0.0849	0.0100	-4.03	-2.89	Yes
AG04	0.3875	0.0868	-3.43	-2.89	Yes
AJ00	0.4601	0.0435	-2.88	-2.89	No
AG01	0.2201	0.1000	-4.25	-2.89	Yes

Table II reports the results of the ADF, KPSS and Phillips–Perron tests. The evidence is mixed across the three procedures, with several SKUs yielding conflicting conclusions regarding stationarity. This is not unexpected, as the tests differ in both their null hypotheses and finite-sample properties. Overall, the results do not provide sufficiently consistent evidence to justify differencing all series prior to nonlinear modelling. Differencing remains available within the ARIMA benchmark, where the objective is purely predictive. Thus, ARIMA is allowed to select $d = 0$ or $d = 1$, while SETAR and MS-AR are estimated on stock levels to preserve inventory-state interpretation. Since the nonlinear models considered in this study are designed to capture nonlinear dynamics, a formal test for nonlinearity is performed before model estimation.

4.3.2 Test for Nonlinear Effects

Before estimating the non-linear forecasting models, a formal test is conducted to assess the presence of non-linear autoregressive dynamics. The Tsay (1986) test was chosen as it provides a general test of non-linear autoregressive alternatives, rather than testing a specific threshold. Following Tsay (1986), the autoregressive order is fixed at $p=2$ for all SKUs to ensure a consistent testing procedure across the series. The null hypothesis is that the data follow a linear autoregressive process, while the alternative provides evidence of nonlinear dynamics.

Table III: Results of the Tsay (1986) Nonlinearity Test

SKU	F statistic	p-value
AA00	3.355	0.0218
AG00	5.252	0.0021
AA03	1.930	0.1295
AA04	3.480	0.0187
AG03	1.747	0.1621
AG04	4.404	0.0059
AJ00	0.560	0.6427
AG01	2.979	0.0350

Note: The null hypothesis is that the series follows a linear autoregressive process.

Table III reports the results of the Tsay (1986) nonlinearity test. The null hypothesis of linearity is rejected for five of the eight SKUs at the 5% significance level, indicating that nonlinear autoregressive dynamics are present in a substantial proportion of the series. While the remaining SKUs do not exhibit statistically significant evidence of nonlinearity under this test, nonlinear forecasting models are estimated for all products to ensure a consistent model comparison framework across the dataset.

4.4 Naïve Benchmark

The naïve benchmark forecasts future stock as equal to the last observed stock level:

$$\hat{X}_{t+h|t} = X_t,$$

This benchmark is included because daily inventory series are highly persistent. Over short horizons, the most recent stock level is often a strong predictor of future stock. Any more complex model should therefore be evaluated against this simple persistence benchmark.

4.5 ARIMA

ARIMA models are included as the main linear benchmark. They capture serial dependence through autoregressive and moving-average terms and can account for non-stationarity through differencing. In compact notation, an $\text{ARIMA}(p, d, q)$ model satisfies

$$\Phi(L)\Delta^d X_t = \Theta(L)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2),$$

where $\Phi(L)$ and $\Theta(L)$ are lag polynomials of order p and q respectively, and $\Delta^d = (1 - L)^d$ is the differencing operator. For each SKU, an ARIMA model is selected automatically using `auto.arima()` with a restricted search space: $p \leq 2$, $q \leq 1$, $d \leq 1$, and `seasonal = FALSE`.

TABLE IV: Selected ARIMA orders and information criteria

SKU	p	d	q	AIC	BIC
AA00	1	0	0	1,683.05	1,691.10
AG00	2	1	1	1,650.24	1,660.93
AA03	2	1	1	1,623.71	1,634.40
AA04	2	1	1	1,586.21	1,596.90
AG03	2	1	1	1,572.82	1,583.51
AG04	2	1	1	1,551.06	1,561.75
AJ00	2	1	0	1,430.20	1,438.22
AG01	2	0	1	1,437.78	1,451.19

Table IV shows that all selected models are relatively simple. Most SKUs are fitted by $\text{ARIMA}(2,1,1)$, while only minor variations occur across products. This reflects the restricted search space adopted in the automatic model selection procedure and provides a consistent linear benchmark for the subsequent forecast comparison. The restrictions are imposed for two reasons. First, the sample contains only 108 daily observations per SKU, so high-order models would increase the risk of overfitting. Second, although daily stock data may contain weekly patterns, the sample spans only around 15 weeks, which is too short to estimate weekly seasonal dynamics reliably. ARIMA therefore serves as a parsimonious linear forecasting benchmark rather than as a fully seasonal model. The selected ARIMA orders are reported in the table below. The notation $\text{ARIMA}(p,d,q)$ indicates the number of autoregressive terms p , the degree

of differencing d , and the number of moving-average terms q . Because the selected order differs across SKUs, the equation above is intentionally written in general form rather than separately substituting the selected p , d and q values for each SKU.

4.6 ETS

ETS models are included as smoothing benchmarks. Unlike ARIMA, which models autocorrelation directly through lagged values and lagged errors, ETS forecasts are based on updating latent components such as level and trend using recent forecast errors. ETS is useful as a benchmark because inventory stock levels often evolve smoothly between adjustment events. For each SKU, the ETS specification is selected automatically using `ets()`, with model choice based on AICc. In practice, the selected models mainly reduce to no-trend and no-seasonality forms, indicating that ETS captures local level persistence rather than systematic trend or seasonal structure.

TABLE V: Selected ETS models and information criteria

SKU	Model	AIC	BIC
AA00	ETS(A,N,N)	1,884.05	1,892.10
AG00	ETS(A,N,N)	1,871.27	1,879.32
AA03	ETS(M,N,N)	1,825.59	1,833.64
AA04	ETS(A,N,N)	1,806.72	1,814.77
AG03	ETS(A,N,N)	1,792.86	1,800.91
AG04	ETS(A,N,N)	1,770.93	1,778.97
AJ00	ETS(A,N,N)	1,644.07	1,652.12
AG01	ETS(A,N,N)	1,642.00	1,650.05

Table V shows that the selected ETS specifications are highly consistent across SKUs. Seven of the eight series are fitted by an ETS(A,N,N) model, indicating additive errors with neither trend nor seasonality, while only AA03 selects an ETS(M,N,N) specification with multiplicative errors. These results suggest that inventory dynamics are primarily characterised by local level persistence rather than systematic trend or seasonal behaviour.

The ETS notation follows the Error–Trend–Seasonality convention, where the first letter denotes the error specification, the second the trend component, and the third the seasonal

component. Thus, ETS(A,N,N) represents additive errors with neither trend nor seasonality, whereas ETS(M,N,N) assumes multiplicative errors. As with ARIMA, the model description is kept general because the selected ETS specification varies across SKUs.

4.7 SETAR

The SETAR model is included to test whether inventory dynamics change when stock crosses an observed threshold. Unlike linear ARIMA models, SETAR allows the autoregressive parameters to differ between regimes. In this study, the threshold variable is a lagged value of the stock series itself, so the model is self-exciting: the past stock level determines which regime applies. A two-regime SETAR model is estimated for each SKU with one threshold, two autoregressive lags, and threshold delay equal to one:

$$X_t = \begin{cases} \alpha_L + \phi_{L,1}X_{t-1} + \phi_{L,2}X_{t-2} + \varepsilon_{L,t}, & X_{t-1} \leq \gamma \\ \alpha_H + \phi_{H,1}X_{t-1} + \phi_{H,2}X_{t-2} + \varepsilon_{H,t}, & X_{t-1} > \gamma \end{cases}$$

The estimated threshold γ separates lower- and higher-inventory regimes. It should not be interpreted as a business reorder point, since operational reorder thresholds are not available. Instead, it is a statistical threshold that identifies where the autoregressive behaviour of the stock series changes. During rolling-origin evaluation, SETAR produced unstable multi-step forecasts for a small number of SKU-horizon combinations. This is a known practical issue with nonlinear threshold models: when forecasts are generated recursively, future regime membership depends on predicted future values, so errors can propagate across horizons. To avoid allowing numerical instability to dominate the comparison, a relative instability filter was applied. If the SETAR RMSE exceeded three times the corresponding naïve RMSE, the SETAR result for that SKU-horizon combination was treated as numerically unstable and excluded from model ranking and average performance summaries.

Table VI reports the estimated threshold for each SKU, the proportion of observations assigned to the lower and higher regimes, and the corresponding in-sample MAPE. The estimated thresholds partition the inventory series into two regimes with varying proportions across SKUs, reflecting differences in inventory dynamics rather than common operating thresholds. The in-sample MAPE values are relatively similar across the sample, indicating comparable in-sample model fit prior to the out-of-sample forecasting exercise.

4.8 MS-AR

The MS-AR model is included as the main latent regime-switching specification. Whereas SETAR assigns regimes using an observed stock threshold, MS-AR assumes that regimes are

unobserved and evolve according to a Markov transition process. This is useful for inventory data because abrupt changes may reflect replenishment, warehouse transfers, or stock reconciliations that are not directly observed in the dataset. For each SKU, a two-regime MS-AR(1) model is estimated:

$$X_t = \alpha_{S_t} + \phi_{S_t} X_{t-1} + \varepsilon_{S_t,t}, \quad \varepsilon_{S_t,t} \sim N(0, \sigma^2_{S_t}),$$

$$p_{ij} = P(S_t = j \mid S_{t-1} = i),$$

where $S_t \in \{1,2\}$ is an unobserved regime following a first-order Markov chain with transition matrix $P = (p_{ij})$. The intercept, autoregressive coefficient, and innovation variance are all allowed to switch across regimes. A two-regime specification is chosen for parsimony and interpretability. Given the short sample length, models with three or more regimes would require estimating many additional parameters and would risk producing regimes with too few observations for reliable inference. The two-regime structure also maps naturally onto the inventory process: one regime can capture normal, persistent stock evolution, while the other can capture short-lived adjustment episodes. It should be noted that regime labels (1 and 2) are arbitrary outputs of the estimation algorithm and vary across SKUs. Regimes are therefore not interpreted by their numerical index, but by their estimated parameters. A regime with an autoregressive coefficient close to one and low innovation variance is characterised as a stable low-volatility inventory evolution regime, while a regime with lower persistence and higher variance is characterised as a volatile adjustment regime. Models are estimated in R using `MSwM::msmFit()` with $k = 2$, $p = 1$, and all three parameters (intercept, AR coefficient, variance) allowed to switch. Multi-step forecasts are generated by propagating the estimated regime probabilities forward through the transition matrix and computing probability-weighted conditional forecasts.

Table VII reports the estimated parameters of the two-regime MS-AR(1) models for each SKU. Although regime labels are arbitrary and therefore not directly comparable across products, a consistent pattern emerges across the sample. For each SKU, one regime is typically characterised by an autoregressive coefficient close to one ($\phi \approx 1$) and a relatively low innovation variance, indicating persistent and stable inventory dynamics, whereas the alternative regime exhibits lower persistence together with substantially higher innovation variance, representing temporary adjustment episodes. Transition probabilities and expected regime durations also vary across SKUs, suggesting that both the persistence and frequency of regime changes are product-specific. Overall, the estimates support the use of a latent regime-

switching specification to capture heterogeneous inventory dynamics across the sample. This heterogeneity further motivates estimating the model separately for each SKU rather than imposing a common regime structure across products.

4.9 Rolling-Origin Forecast Evaluation

Forecast performance is evaluated using an expanding-window rolling-origin design. Random train-test splitting is not appropriate for time series because it would break the chronological order of the data and allow future information to influence model estimation. Instead, each model is estimated using only observations available up to a given forecast origin and evaluated on future observations. The initial training window contains $n_{init} = 80$ observations. For each forecast horizon $h \in \{1, 7, 14\}$, the forecast origin moves forward one day at a time. At each origin, the model is re-estimated on all observations available up to that date and an h -step-ahead forecast is produced. This expanding-window design mimics a real operational forecasting setting in which new data become available over time and the model is updated sequentially. The horizons $h = 1, 7, 14$ were chosen to represent next-day, one-week, and two-week inventory planning windows.

Figure 3: Monte Carlo forecasting workflow for inventory risk estimation.

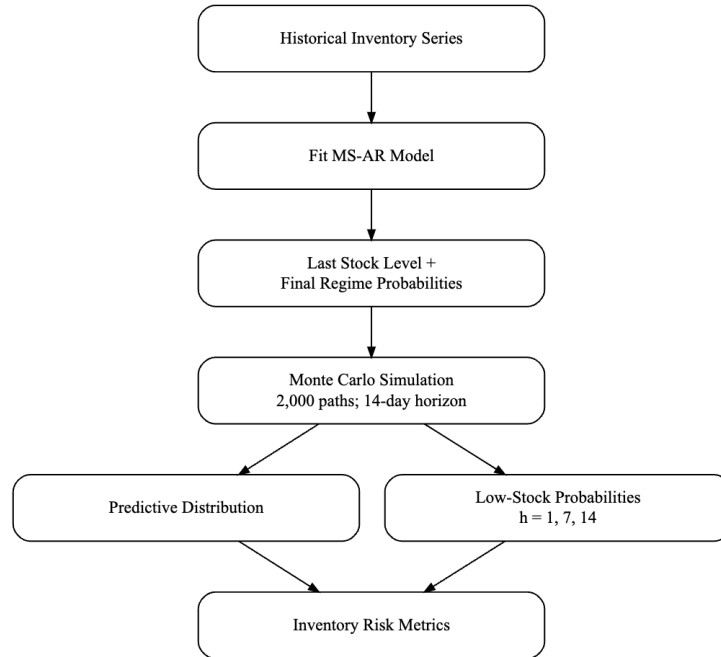


Figure 3 summarises the Monte Carlo forecasting workflow used to generate predictive inventory distributions and low-stock probabilities from the estimated MS-AR model.

4.10 Forecast Accuracy Metrics

Forecast accuracy is evaluated using the Root Mean Squared Error (RMSE) and the Mean Absolute Scaled Error (MASE). Forecast errors are first defined as

$$e_{i,h} = X_{t_i+h} - \hat{X}_{t_i+h|t_i},$$

where $e_{i,h}$ denotes the forecast error for forecast origin t_i and horizon h . The Root Mean Squared Error is defined as

$$RMSE(h) = \sqrt{\frac{1}{N_h} \sum_{i=1}^{N_h} e_{i,h}^2},$$

where N_h denotes the number of valid forecast origins for horizon h . RMSE is expressed in inventory units and gives greater weight to large forecast errors because errors are squared before averaging. The Mean Absolute Scaled Error is defined as

$$MASE(h) = \frac{\frac{1}{N_h} \sum_{i=1}^{N_h} |e_{i,h}|}{\frac{1}{n_{init}-1} \sum_{t=2}^{n_{init}} |X_t - X_{t-1}|},$$

where n_{init} denotes the length of the initial training sample used to compute the scaling factor. MASE scales the model's mean absolute forecast error by the average one-step naïve error on the initial training sample. This makes the metric comparable across SKUs with different stock scales. A MASE value below one indicates that the model improves upon the naïve benchmark in absolute-error terms.

4.11 SETAR Forecast Stability Filter

SETAR forecasts are screened using a stability filter designed to exclude degenerate predictions.

$$RMSE_{SETAR} > 3 RMSE_{Naïve}$$

For each combination of SKU and forecast horizon, forecasts that meet the criterion above are discarded and excluded from the model rankings and average performance summaries.

$$|e_{SETAR}| > 3|e_{Naïve}|$$

At the individual forecast level, observations satisfying this condition are removed before constructing the loss matrix used in the Model Confidence Set (MCS) procedure. The multiplier of three is an empirical threshold chosen to identify clearly unstable forecasts while retaining ordinary forecast variability.

4.12 Statistical Forecast Comparison

Average forecast accuracy measures do not indicate whether performance differences are statistically significant. Therefore, two statistical comparison procedures are applied. First, Diebold–Mariano test is applied pairwise to compare MS-AR with each benchmark model. Squared forecast errors are used as the loss function, so the loss differential is defined as

$$d_t = e^2_{MSAR,t} - e^2_{benchmark,t},$$

where d_t denotes the squared-loss differential between MS-AR and the competing benchmark at time t . The null hypothesis is equal predictive accuracy, $E(d_t)=0$, while the one-sided alternative is $E(d_t)<0$, indicating lower expected squared forecast loss for MS-AR. Because the evaluation includes multi-step rolling-origin forecasts, loss differentials may be serially correlated, particularly at the 7-day and 14-day horizons. To account for this dependence, the Bartlett long-run variance estimator is used when computing the test statistic. This provides a more stable estimate of the variance of the loss differential in short samples and avoids unstable variance estimates that can arise in overlapping multi-step forecast comparisons. Second, the Model Confidence Set (MCS) procedure is applied to evaluate all competing models jointly. Unlike the DM test, which is pairwise, the MCS identifies the set of models that are not statistically inferior to the best model at a chosen significance level. The procedure is implemented with $\alpha = 0.10$, the $T_{\max}$ statistic, 5,000 bootstrap replications, and squared forecast errors as the loss function

4.13 Low-Stock Risk Simulation

To translate the fitted MS-AR model into an operational risk indicator, a Monte Carlo simulation is performed for each SKU. The objective is to estimate the probability that the inventory level falls below a predefined low-stock threshold at different forecast horizons.

Figure 4: Illustration of the Monte Carlo low-stock risk simulation framework

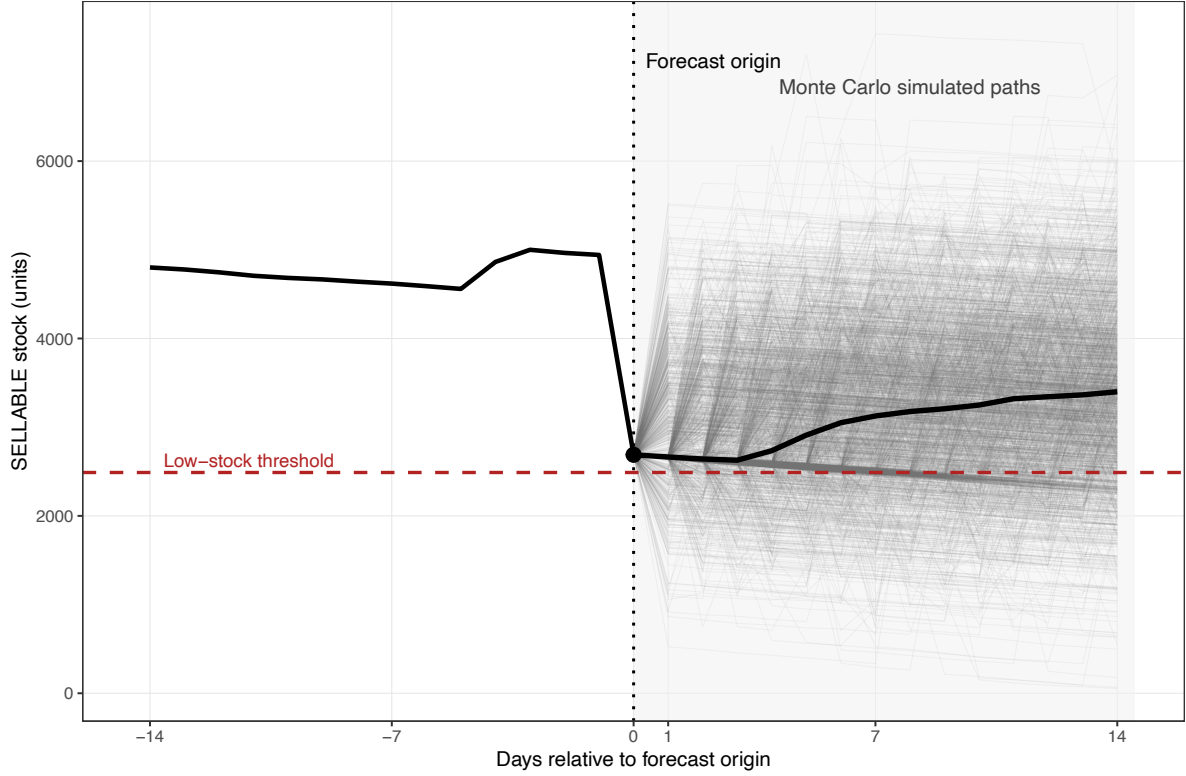


Figure 4 illustrates the simulation framework for one representative SKU (AA03). The simulation starts from the last observed inventory level and generates 2,000 possible future inventory trajectories over a 14-day forecast horizon. Each grey line represents one simulated future path, while the solid black line shows the median simulated trajectory. The dashed red line indicates the low-stock threshold used to evaluate inventory risk.

For each SKU, the MS-AR model is re-estimated using the full sample. Starting from the last observed stock level and the final smoothed regime probabilities, future inventory paths are generated recursively. At each step, the active regime is sampled according to the estimated transition matrix, and the next inventory level is computed using the corresponding regime-specific intercept, autoregressive coefficient, and innovation variance. The probability of low stock is then estimated at the three forecast horizons $h\{1,7,14\}$ as the proportion of simulated paths for which the inventory level at horizon h lies below the predefined threshold.

Because operational reorder points and safety-stock levels are not available, the threshold is defined statistically as the 10th percentile of the historical inventory distribution for each SKU. This threshold should therefore be interpreted as a relative low-stock indicator, rather than as an actual stockout or replenishment threshold. Finally, the analysis is repeated using the 5th and 20th percentiles to assess the sensitivity of the estimated probabilities to the threshold definition. This robustness analysis evaluates whether the conclusions remain stable under alternative definitions of low inventory.

4.14 Software and R Packages

All empirical analyses were conducted in R. Data preparation and manipulation relied on readxl, dplyr, lubridate, purrr, tibble and tidyr; graphical analysis used ggplot2; time-series modelling and diagnostic procedures used forecast, tseries, urca and tsDyn; Markov-switching models were estimated using MSwM; the Model Confidence Set procedure used MCS; and knitr and kableExtra were used to prepare and format analytical outputs.

5 RESULTS

5.1 Estimated Regime Structure

5.1.1 Markov-Switching Autoregressive Model

Table VII reports the estimated parameters of the MS-AR(1) model for each of the eight selected Master SKUs. For each stock series, the model identifies two latent regimes characterised by distinct intercepts, autoregressive coefficients, conditional standard deviations, self-transition probabilities, and implied expected durations.

TABLE VII: MS-AR(1) estimated parameters by SKU

SKU	α_1	ϕ_1	σ_1	p_{11}	Dur. ₁	α_2	ϕ_2	σ_2	p_{22}	Dur. ₂
AA00	-44.0	0.999	14.7	0.766	4.3	2,781.5	0.558	917.9	0.478	1.9
AG00	7,528.3	-0.655	844.1	0.084	1.1	10.9	0.993	35.3	0.879	8.3
AA03	1,823.7	0.505	838.4	0.300	1.4	-31.5	1.002	8.4	0.771	4.4
AA04	2,919.2	0.088	678.3	0.131	1.2	-13.4	0.998	7.2	0.754	4.1
AG03	0.4	0.996	4.3	0.756	4.1	2,011.5	0.304	621.5	0.315	1.5
AG04	2,657.6	0.027	550.4	0.404	1.7	0.6	0.996	4.2	0.828	5.8
AJ00	-14.1	1.001	4.7	0.820	5.6	1,678.1	0.242	348.5	0.279	1.4
AG01	1,284.4	0.107	321.6	0.268	1.4	0.0	0.995	3.7	0.805	5.1

Table VII shows a consistent two-regime pattern across the selected Master SKUs, although the numerical labels of the regimes are not directly comparable across products. For each series, one regime is characterised by an autoregressive coefficient close to one, relatively low innovation variance, and a longer expected duration. This pattern is consistent with stable and highly persistent inventory evolution. The alternative regime generally shows lower autoregressive persistence, substantially higher innovation variance, and a shorter expected duration, indicating a more volatile adjustment state.

Regime labels in MS-AR models are not identified a priori. The assignment of “Regime 1” or “Regime 2” is arbitrary and varies across SKUs. Regimes are therefore interpreted on the basis of their estimated parameters rather than their numerical labels. Despite differences in inventory scale, the distinction between a persistent low-volatility regime and a shorter-lived

high-volatility regime appears consistently across the eight series. The available stock data do not identify the operational events responsible for these changes, but the volatile regimes may be associated with replenishment activities, inventory transfers, reconciliations, or other stock adjustments.

5.1.2 Self-Exciting Threshold Autoregressive Model

TABLE VIII: SETAR estimated thresholds, regime proportions and in-sample MAPE

SKU	Threshold	% Low regime	% High regime	In-sample MAPE (%)
AA00	4,855	15.1	84.9	4.87
AG00	5,304	71.7	28.3	6.17
AA03	4,863	84.9	15.1	6.11
AA04	2,830	25.5	74.5	6.43
AG03	3,397	78.3	21.7	6.99
AG04	3,072	71.7	28.3	6.94
AJ00	1,997	26.4	73.6	4.47
AG01	1,562	63.2	36.8	7.12

Table VIII reports the estimated SETAR thresholds, regime proportions, and in-sample MAPE for each Master SKU. Estimated thresholds vary substantially across SKUs, ranging from 1,562 units for AG01 to 5,304 units for AG00. This variation reflects differences in the scale of inventory held across products. The SETAR thresholds should therefore be interpreted as SKU-specific relative inventory-state thresholds rather than common operational stockout or reorder points.

The proportion of observations in each regime also varies considerably. AA03 and AG03 spend approximately 79-85% of the sample period below the estimated threshold, indicating that these SKUs operate predominantly in the lower-inventory regime. In contrast, AA00 and AA04 spend only around 15-26% of observations below the threshold, suggesting that their stock levels are usually maintained above the switching level. Other SKUs display more balanced but still asymmetric regime allocations. In-sample fit, measured by MAPE, ranges from approximately 4.47% for AJ00 to 7.12% for AG01. The relatively similar MAPE values indicate comparable in-sample fit across the selected SKUs. However, the interpretation of

regime-specific coefficients should remain cautious, particularly for SKUs with highly imbalanced regime proportions, since a small number of observations in one regime may lead to less stable parameter estimates.

5.2 Forecast Accuracy

Table IX: Average RMSE by forecast horizon

Horizon	ARIMA	ETS	SETAR	MS-AR	Naive	Best
1	767.0	745.8	969.3	721.5	741.8	MS-AR
7	931.6	1,090.4	1,248.7	848.9	982.0	MS-AR
14	1,091.8	1,664.8	1,002.4	823.9	1,201.0	MS-AR

Table IX reports the average RMSE across SKUs at each forecast horizon, while Table X reports the number of SKU-horizon combinations in which each model achieves the lowest RMSE or MASE.

TABLE X: Win counts across SKU-horizon combinations

Model	RMSE wins	MASE wins
MS-AR	17	6
ARIMA	4	3
SETAR	2	1
ETS	1	3
Naive	0	11

The win-count procedure is included as a descriptive summary of forecast accuracy rankings, not as a formal statistical test. Its purpose is to show whether good model performance is broad-based across the forecasting exercise or concentrated in only a few cases. The evaluation contains 24 SKU-horizon cases, obtained from 8 SKUs and 3 forecast horizons: $8 \times 3 = 24$. For each case, the model with the lowest RMSE receives one RMSE win, and the model with the lowest MASE receives one MASE win. For example, AA00 at $h = 1$, AA00 at $h = 7$, and AA00 at $h = 14$ are three separate cases. Table X is useful because average RMSE or MASE values can hide heterogeneity across SKUs and horizons. A model may achieve a low average error because it performs very well in a small number of high-scale or difficult cases, while performing less consistently elsewhere. The win count therefore complements average accuracy

metrics by showing how often each model ranks first across the 24 cases. It should be interpreted as an intuitive measure of consistency in relative forecast performance. Formal statistical evidence is provided separately by the Diebold-Mariano tests and the Model Confidence Set procedure. In terms of RMSE, MS-AR records the largest number of RMSE wins, with 17 wins out of 24 SKU-horizon cases. The remaining RMSE wins are distributed as follows: ARIMA with 4 wins, SETAR with 2 wins, ETS with 1 win, Naive with 0 wins. When averaged across SKUs by forecast horizon, MS-AR also produces the lowest mean RMSE at all three horizons. This indicates that MS-AR performs particularly well under a loss measure that gives greater weight to large forecast errors, especially at the longer horizons considered in the analysis. The MASE results provide a more nuanced picture. Naive records the largest number of MASE wins overall, with 11 wins. The remaining MASE wins are distributed as follows: MS-AR with 6 wins, ARIMA with 3 wins, ETS with 3 wins, SETAR with 1 win. The strong performance of the naïve model under MASE is consistent with the high persistence observed in the inventory series, since the most recent stock level provides a difficult benchmark to outperform. At the same time, Table IX shows that MS-AR becomes relatively more competitive as the forecast horizon increases. The divergence between RMSE and MASE rankings is informative. RMSE penalises large forecast errors quadratically and is therefore more sensitive to occasional large stock movements, such as replenishment events or inventory adjustments. The strong RMSE performance of MS-AR indicates that the model is comparatively less affected by large forecast errors in several SKU-horizon combinations. This result is consistent with, but does not by itself prove, the idea that allowing the dynamics to change across regimes can be useful when the stock series contains abrupt movements. MASE, by contrast, evaluates forecast errors relative to the in-sample naïve benchmark and therefore highlights the difficulty of improving upon simple persistence in highly persistent inventory series, especially at short horizons.

5.3 Diebold-Mariano Tests

Table XI: Significant Diebold-Mariano results

SKU	h	Comparison	DM statistic	p-value
AA00	1	MSAR vs ETS	-2.566	0.0081
AG04	1	MSAR vs ARIMA	-1.902	0.0340
AG04	1	MSAR vs ETS	-2.467	0.0101

Table XI reports the statistically significant Diebold-Mariano comparisons between MS-AR and the benchmark models. The null hypothesis is equal predictive accuracy, while the one-sided alternative tests whether MS-AR has lower squared forecast loss. Pairwise Diebold-Mariano tests were conducted using squared forecast errors, consistent with the RMSE evaluation criterion, and a Bartlett long-run variance estimator was used to account for serial correlation in multi-step loss differentials. The results provide limited statistical evidence of superior predictive accuracy. At the one-day horizon, MS-AR significantly outperforms ARIMA for one SKU and ETS for two SKUs. No statistically significant advantage is detected against the naïve benchmark or SETAR at any horizon. At the seven- and fourteen-day horizons, no pairwise comparison reaches statistical significance. These findings should be interpreted in the context of the short evaluation samples and the dependence present in multi-step forecast errors, which reduce the ability to detect small differences in predictive accuracy. MS-AR remains the best-performing specification in terms of RMSE wins, but statistically significant pairwise dominance cannot be claimed uniformly across SKUs and horizons.

5.4 Model Confidence Set

Table XII reports the Model Confidence Set inclusion rates across the 23 valid SKU-horizon combinations.

Table XII: Model Confidence Set inclusion rates

Model	Included / Total	Inclusion rate
MS-AR	23/23	100%
SETAR	21/23	91.3%
ARIMA	15/23	65.2%
ETS	13/23	56.5%
Naive	13/23	56.5%

The Model Confidence Set procedure of Hansen, Lunde and Nason (2011) was applied to identify, for each SKU-horizon combination, the set of models that are not statistically worse than the best-performing model at a significance level of $\alpha = 0.10$. The procedure used the $T_{\max}$ statistic with 5,000 bootstrap replications and squared forecast errors as the loss criterion, consistent with the RMSE evaluation. MS-AR is the only model never eliminated from the superior set across the valid SKU-horizon combinations. SETAR is the second most frequently included model, followed by ARIMA, while ETS and the naïve benchmark have the same inclusion rate. The MCS results complement rather than replicate the Diebold-Mariano findings. While the DM tests provide limited evidence of statistically significant pairwise superiority, the MCS shows that MS-AR is consistently competitive and is never eliminated from the superior set across the valid SKU-horizon combinations. However, this does not imply that MS-AR is uniquely superior in every case, since several competing models also remain in the superior set for many SKU-horizon combinations.

5.5 Low-Stock Risk

Table XIII reports the estimated probability that inventory is below the SKU-specific relative low-stock threshold at forecast horizons of 1, 7, and 14 days.

Table XIII: Monte Carlo low-stock risk probabilities by forecast horizon

SKU	Current stock	Threshold (10th pct.)	h = 1	h = 7	h = 14
AA00	3,961	4,645	0.690	0.262	0.150
AG00	2,896	4,307	0.925	0.581	0.470
AA03	2,689	2,489	0.072	0.226	0.203
AA04	2,270	2,648	0.901	0.390	0.258
AG03	2,092	2,260	0.774	0.302	0.194
AG04	1,500	2,196	0.670	0.280	0.208
AJ00	1,850	1,751	0.040	0.124	0.172
AG01	1,009	1,173	0.794	0.380	0.264

To complement the point forecast evaluation, a Monte Carlo simulation was conducted for each Master SKU using the fitted MS-AR model. Starting from the last observed stock level and the final smoothed regime probability vector, 2,000 future inventory paths were simulated over a 14-day horizon. At each step, the regime was sampled from the estimated transition probabilities, and the next stock level was generated using the corresponding regime-specific parameters. For each forecast horizon $h \in \{1, 7, 14\}$, the low-stock probability is calculated as the proportion of simulated paths for which the stock level is below the SKU-specific threshold exactly at that horizon. It should therefore not be interpreted as the probability of entering low stock at any point within the forecast horizon.

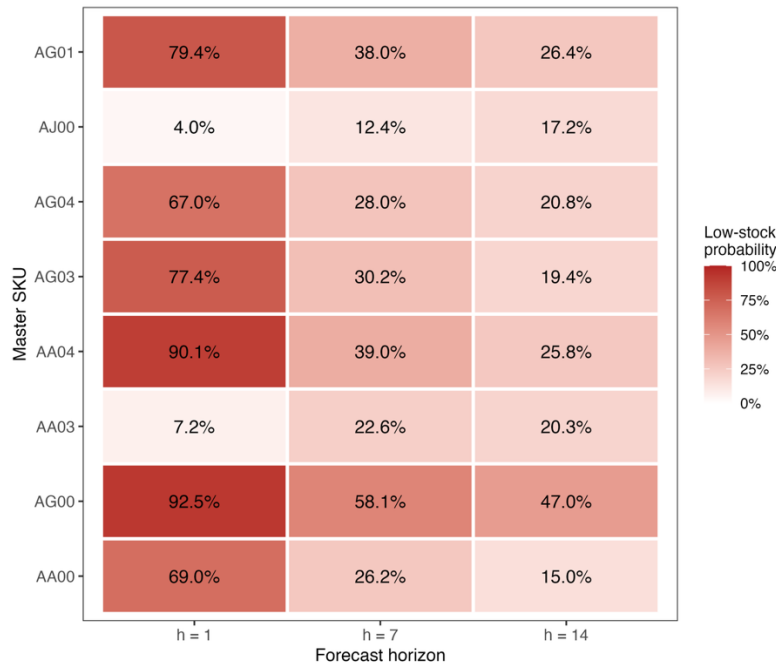
Because operational reorder points and safety-stock levels are not available, the threshold is defined as the 10th percentile of each SKU's historical stock distribution. It identifies the lower tail of the observed inventory range and should therefore be interpreted as a relative low-stock indicator rather than as an operational reorder point or physical stockout threshold.

Table XIII shows substantial differences across Master SKUs. Several products have a high probability of being below their relative threshold at the one-day horizon, while AA03 and AJ00 show much lower short-term probabilities. For several SKUs, the estimated probability decreases at longer horizons. This does not imply that low-stock risk generally decreases over time. The simulations are conditional on the latest observed stock level and reproduce the dynamics estimated from the historical inventory series. Since those historical dynamics

include both gradual stock movements and upward or downward adjustments, the predicted distribution may move away from the low-stock region at longer horizons. The results should therefore be interpreted as horizon-specific conditional probabilities under the fitted MS-AR dynamics, rather than as a simulation of what would happen if no replenishment action were taken. The model does not separately observe or simulate replenishment decisions.

Figure XIV provides a visual summary of the probabilities reported in Table XIII. Darker colours indicate a higher estimated probability that inventory is below the SKU-specific relative low-stock threshold at the corresponding forecast horizon.

Figure 5: Heatmap of estimated conditional probabilities that inventory is below the SKU-specific relative low-stock threshold at forecast horizons of 1, 7 and 14 days.



The heatmap highlights substantial heterogeneity in the estimated low-stock probabilities across Master SKUs. Products such as AG00, AA04, AG03 and AG01 exhibit relatively high short-term probabilities of remaining below their historical low-stock threshold, whereas AA03 and AJ00 display considerably lower probabilities at the one-day horizon. For several SKUs, the estimated probability decreases as the forecast horizon increases. This pattern should not be interpreted as evidence that inventory risk naturally declines over time. Rather, the probabilities are conditional on the fitted MS-AR dynamics and the latest observed stock level. Since the historical inventory process includes both gradual stock evolution and upward or downward adjustments, the simulated distribution may move away from the low-stock region at longer horizons.

The heatmap therefore complements Table XIII by providing an immediate visual comparison of the relative low-stock risk across products and forecast horizons, while preserving the interpretation of the probabilities as conditional model-based estimates rather than operational stockout probabilities.

6 DISCUSSION AND CONCLUSION

This study examined whether daily Amazon FBA inventory stock series exhibit regime-switching dynamics and whether explicitly modelling such dynamics improves forecast accuracy relative to linear benchmarks. The results provide consistent evidence in support of regime-dependent inventory behaviour, while also revealing important nuances regarding the statistical significance and practical scope of the forecasting gains.

The most robust empirical finding is the consistency of the two-regime structure identified by the MS-AR model across all eight selected SKUs. Despite substantial heterogeneity in stock scale and product characteristics, every series displays one regime characterised by high persistence and relatively low innovation variance, together with a second regime displaying lower persistence and substantially higher volatility. This regularity is consistent with the observed behaviour of the inventory series, which alternates between periods of gradual stock evolution and shorter episodes of more pronounced adjustment. The available data do not identify the operational events generating these changes, but they are compatible with replenishment activities, inventory transfers, inventory reconciliations, or other warehouse operations. The SETAR model supports the presence of nonlinear dynamics from a different perspective by identifying SKU-specific stock thresholds separating two regimes, although the interpretation of threshold regimes remains sensitive to regime imbalance.

In terms of forecast performance, MS-AR achieves the strongest overall RMSE performance and produces the lowest average RMSE at all three forecast horizons. At the one-day horizon, however, the naïve persistence benchmark remains highly competitive under MASE, reflecting the strong persistence of daily inventory series. This finding is consistent with the broader forecasting literature, where simple benchmarks frequently match or outperform more complex models at short horizons when the underlying process is highly persistent.

The Diebold-Mariano tests and the Model Confidence Set provide complementary evidence. The DM tests yield only limited evidence of statistically significant pairwise superiority, which

is consistent with the relatively short evaluation samples and the limited power of pairwise forecast comparison tests. By contrast, the MCS evaluates all competing models jointly and shows that MS-AR is never eliminated from the superior set across the valid SKU-horizon combinations. This should not be interpreted as evidence that MS-AR is uniquely superior in every case, since several competing models also remain in the superior set. Rather, the combined evidence indicates that MS-AR is consistently competitive without establishing uniform statistical dominance over all benchmark models.

The Monte Carlo simulation extends the forecasting exercise beyond point predictions by providing horizon-specific probabilities that inventory is below a relative low-stock threshold. These probabilities are conditional on the fitted MS-AR dynamics and the latest observed stock level. They should not be interpreted as the probability of entering a low-stock state within the forecast horizon, nor as the effect of explicit replenishment decisions. Instead, they summarise the distribution of future inventory levels implied by the estimated regime-switching process. For several SKUs, the estimated probability decreases at longer forecast horizons. This pattern should not be interpreted as evidence that inventory risk naturally declines over time. Rather, it reflects the behaviour of the fitted statistical model, whose simulated trajectories reproduce the historical inventory dynamics observed in the data. The proposed framework therefore complements traditional point forecasts with probabilistic information about future inventory states, while remaining conditional on the observed inventory process.

Several limitations should be acknowledged. First, the rolling-origin evaluation is conducted on relatively short series of approximately 108 daily observations per SKU, which constrains both the length of the training window and the number of evaluation origins available for statistical inference. Although this time dimension is sufficient for a parsimonious forecasting comparison, it remains limited for estimating regime-switching models by maximum likelihood. For this reason, the analysis restricts the MS-AR specification to two regimes and one autoregressive lag, evaluates performance through rolling-origin out-of-sample validation, and interprets regime estimates cautiously. Second, the Monte Carlo simulation assumes Gaussian innovations within each regime, consistent with the parametric assumptions of the MS-AR model, but does not account for possible heavy-tailed or asymmetric shock distributions. Third, the low-stock threshold is a statistical construct based on historical percentiles and does not correspond to operational concepts such as reorder points or safety-stock levels, which were not available for this analysis. Fourth, the analysis models aggregate

inventory summed across warehouse locations, which is appropriate for product-level inventory availability but may obscure location-specific dynamics relevant to fulfilment performance. Finally, the observed inventory series combine the effects of demand, replenishment activities, inventory transfers, and other operational adjustments into a single stock process. Since these components are not observed separately, the model captures their combined statistical behaviour rather than the underlying inventory mechanism. Consequently, the estimated regimes should be interpreted as latent statistical states rather than direct representations of specific warehouse operations.

Several directions emerge naturally from these findings. Longer time series would allow richer seasonal structures, more flexible regime-switching specifications, and stronger statistical inference. Future work could also compare aggregate Master SKU models with SKU-location models or partially pooled specifications, allowing warehouse-level heterogeneity to be analysed without sacrificing estimation stability. Incorporating exogenous information, such as promotional activity, seasonality indicators, inbound shipment data, lead times, or sales information, into the switching mechanism could improve both forecast accuracy and regime interpretability. Extending the analysis to a larger cross-section of products would allow formal testing of the cross-SKU regularity observed in this study. A particularly promising direction would be to model the inventory balance explicitly by separating inventory depletion from replenishment events, rather than forecasting the observed stock series alone. Such an approach could distinguish demand-driven inventory changes from operational replenishment decisions, providing a richer representation of warehouse dynamics. Future research could also investigate alternative probabilistic risk measures, such as the probability of entering a low-stock state within a forecast horizon or the expected time until a threshold is first crossed, which may provide more direct support for inventory planning decisions. Finally, integrating operational reorder points or safety-stock targets into the simulation framework would strengthen the connection between statistical forecasting and inventory management.

Overall, this thesis shows that regime-switching models, and MS-AR in particular, provide a flexible statistical framework for modelling daily Amazon FBA inventory dynamics. Their principal contribution is not that they uniformly outperform simpler forecasting benchmarks, but that they combine competitive point forecast performance with an interpretable description of changing inventory dynamics and horizon-specific probabilistic low-stock risk estimates. In this sense, regime-switching models should be viewed as a complement to traditional

forecasting approaches rather than a replacement, particularly when uncertainty surrounding future inventory levels is of practical interest.

APPENDIX

Detailed RMSE Results

Table XIV: Detailed rolling-origin RMSE by SKU, horizon, and model

SKU	Horizon	ARIMA	ETS	SETAR	MS-AR	Naive	Best model
AA00	1	1,041.23	1,051.81	1,053.56	1,032.31	1,051.13	MSAR
AA00	7	1,232.71	1,403.28	1,155.36	1,149.32	1,430.62	MSAR
AA00	14	972.83	1,591.42	1,007.40	974.33	1,630.65	ARIMA
AG00	1	1,107.15	1,059.40	1,558.95	1,050.63	1,057.78	MSAR
AG00	7	1,401.94	1,392.64	2,971.92	1,068.61	1,430.92	MSAR
AG00	14	1,760.29	1,744.50	1,519.23	877.33	1,789.25	MSAR
AA03	1	921.84	908.61	941.96	879.98	895.63	MSAR
AA03	7	1,072.13	2,031.40	1,158.05	1,184.19	1,145.86	ARIMA
AA03	14	1,250.03	4,853.96	1,282.57	1,313.60	1,327.90	ARIMA
AA04	1	836.49	844.67	865.41	764.55	800.43	MSAR
AA04	7	1,005.45	1,049.10	881.87	913.22	1,032.60	SETAR
AA04	14	1,288.75	1,366.58	917.20	948.82	1,309.42	SETAR
AG03	1	782.07	730.35		712.92	745.34	MSAR
AG03	7	940.07	937.03		873.62	968.16	MSAR
AG03	14	1,236.21	1,195.64		892.66	1,243.53	MSAR
AG04	1	695.44	663.28	1,202.22	636.53	662.50	MSAR

SKU	Horizon	ARIMA	ETS	SETAR	MS-AR	Naive	Best model
AG04	7	849.67	851.43	796.88	682.10	874.98	MSAR
AG04	14	1,027.36	994.66	657.09	558.89	1,046.35	MSAR
AJ00	1	349.76	338.43	788.00	340.45	352.52	ETS
AJ00	7	447.82	449.38	528.04	478.39	468.42	ARIMA
AJ00	14	565.62	580.40	630.82	561.10	610.69	MSAR
AG01	1	401.97	369.93	374.69	354.33	369.33	MSAR
AG01	7	502.99	608.63		442.05	504.43	MSAR
AG01	14	633.60	991.52		464.65	649.96	MSAR

Detailed MASE Results

TABLE XV: Detailed rolling-origin MASE by SKU, horizon, and model

SKU	Horizon	ARIMA	ETS	SETAR	MS-AR	Naive	Best model
AA00	1	3.783	3.852	3.876	3.691	3.667	NAIVE
AA00	7	5.953	7.262	6.082	5.795	7.348	MSAR
AA00	14	5.268	10.048	6.293	5.704	10.563	ARIMA
AG00	1	9.700	8.700	14.241	9.943	7.934	NAIVE
AG00	7	15.622	15.208	23.452	15.494	15.049	NAIVE
AG00	14	24.765	24.222	14.754	12.910	24.099	MSAR
AA03	1	5.920	5.737	6.035	5.533	5.005	NAIVE
AA03	7	9.420	16.239	10.964	11.974	9.855	ARIMA
AA03	14	12.807	34.217	14.056	15.407	13.367	ARIMA
AA04	1	9.305	9.529	10.103	8.657	7.679	NAIVE
AA04	7	14.300	15.949	14.987	16.567	13.991	NAIVE
AA04	14	22.033	23.047	16.787	17.909	21.455	SETAR
AG03	1	10.774	9.601		10.207	8.597	NAIVE

SKU	Horizon	ARIMA	ETS	SETAR	MS-AR	Naive	Best model
AG03	7	17.202	16.651		19.693	17.069	ETS
AG03	14	26.314	25.210		20.449	25.436	MSAR
AG04	1	10.438	9.555	15.079	8.852	8.283	NAIVE
AG04	7	16.509	16.459	17.167	13.883	16.494	MSAR
AG04	14	24.935	23.843	16.640	13.264	24.185	MSAR
AJ00	1	5.098	4.477	8.438	4.866	4.203	NAIVE
AJ00	7	9.191	7.988	10.818	10.916	8.450	ETS
AJ00	14	13.225	12.444	14.004	13.748	13.477	ETS
AG01	1	9.786	8.478	8.414	8.184	6.922	NAIVE
AG01	7	16.037	18.014		16.043	13.949	NAIVE
AG01	14	22.412	31.619		17.962	22.096	MSAR

Detailed Diebold-Mariano Tests

Table XVI: Full Diebold-Mariano test results

SKU	Horizon	Comparison	DM statistic	p-value	n	Significant
AA00	1	MSAR vs ARIMA	-0.821	0.2093	28	false
AA00	1	MSAR vs ETS	-2.566	0.0081	28	true
AA00	1	MSAR vs SETAR	-0.219	0.4142	23	false
AA00	1	MSAR vs NAIVE	-1.668	0.0535	28	false
AA00	7	MSAR vs ARIMA	-0.937	0.1796	22	false
AA00	7	MSAR vs ETS	-1.585	0.0639	22	false
AA00	7	MSAR vs SETAR	0.164	0.5640	18	false
AA00	7	MSAR vs NAIVE	-1.505	0.0736	22	false
AA00	14	MSAR vs ARIMA	0.002	0.5006	15	false
AA00	14	MSAR vs ETS	-0.521	0.3052	15	false
AA00	14	MSAR vs SETAR	-0.056	0.4781	15	false
AA00	14	MSAR vs NAIVE	-0.613	0.2748	15	false
AG00	1	MSAR vs ARIMA	-1.633	0.0575	26	false

SKU	Horizon	Comparison	DM statistic	p-value	n	Significant
AG00	1	MSAR vs ETS	-1.459	0.0785	26	false
AG00	1	MSAR vs SETAR	-0.653	0.2622	15	false
AG00	1	MSAR vs NAIVE	-1.256	0.1104	26	false
AG00	7	MSAR vs ARIMA	-0.917	0.1853	20	false
AG00	7	MSAR vs ETS	-0.881	0.1946	20	false
AG00	7	MSAR vs SETAR	0.157	0.5609	13	false
AG00	7	MSAR vs NAIVE	-0.889	0.1925	20	false
AG00	14	MSAR vs ARIMA			11	false
AG00	14	MSAR vs ETS			11	false
AG00	14	MSAR vs SETAR			9	false
AG00	14	MSAR vs NAIVE			11	false
AA03	1	MSAR vs ARIMA	-1.341	0.0956	28	false
AA03	1	MSAR vs ETS	-1.348	0.0944	28	false
AA03	1	MSAR vs SETAR	-1.284	0.1073	20	false
AA03	1	MSAR vs NAIVE	-1.001	0.1629	28	false
AA03	7	MSAR vs ARIMA	1.919	0.9657	22	false
AA03	7	MSAR vs ETS	-0.833	0.2070	22	false
AA03	7	MSAR vs SETAR	0.624	0.7296	19	false
AA03	7	MSAR vs NAIVE	0.452	0.6722	22	false
AA03	14	MSAR vs ARIMA	0.120	0.5467	15	false
AA03	14	MSAR vs ETS	-0.169	0.4341	15	false
AA03	14	MSAR vs SETAR			12	false
AA03	14	MSAR vs NAIVE	-0.029	0.4886	15	false
AA04	1	MSAR vs ARIMA	-1.674	0.0529	28	false
AA04	1	MSAR vs ETS	-1.471	0.0765	28	false
AA04	1	MSAR vs SETAR	-0.319	0.3769	16	false
AA04	1	MSAR vs NAIVE	-1.540	0.0676	28	false
AA04	7	MSAR vs ARIMA	-0.587	0.2817	22	false
AA04	7	MSAR vs ETS	-1.020	0.1596	22	false

SKU	Horizon	Comparison	DM statistic	p-value	n	Significant
AA04	7	MSAR vs SETAR	0.052	0.5201	12	false
AA04	7	MSAR vs NAIVE	-0.653	0.2604	22	false
AA04	14	MSAR vs ARIMA	-0.357	0.3632	15	false
AA04	14	MSAR vs ETS	-0.745	0.2342	15	false
AA04	14	MSAR vs SETAR			12	false
AA04	14	MSAR vs NAIVE	-0.355	0.3638	15	false
AG03	1	MSAR vs ARIMA	-1.691	0.0511	28	false
AG03	1	MSAR vs ETS	-1.071	0.1468	28	false
AG03	1	MSAR vs SETAR	-1.038	0.1590	14	false
AG03	1	MSAR vs NAIVE	-1.335	0.0966	28	false
AG03	7	MSAR vs ARIMA	-0.363	0.3602	22	false
AG03	7	MSAR vs ETS	-0.340	0.3688	22	false
AG03	7	MSAR vs SETAR	0.184	0.5717	16	false
AG03	7	MSAR vs NAIVE	-0.460	0.3251	22	false
AG03	14	MSAR vs ARIMA	-0.304	0.3828	15	false
AG03	14	MSAR vs ETS	-0.285	0.3898	15	false
AG03	14	MSAR vs SETAR			13	false
AG03	14	MSAR vs NAIVE	-0.278	0.3927	15	false
AG04	1	MSAR vs ARIMA	-1.902	0.0340	28	true
AG04	1	MSAR vs ETS	-2.467	0.0101	28	true
AG04	1	MSAR vs SETAR	-0.637	0.2679	13	false
AG04	1	MSAR vs NAIVE	-1.692	0.0511	28	false
AG04	7	MSAR vs ARIMA	-1.307	0.1027	22	false
AG04	7	MSAR vs ETS	-1.302	0.1035	22	false
AG04	7	MSAR vs SETAR	-1.103	0.1426	18	false
AG04	7	MSAR vs NAIVE	-1.241	0.1141	22	false
AG04	14	MSAR vs ARIMA	-0.605	0.2774	15	false
AG04	14	MSAR vs ETS	-0.573	0.2880	15	false
AG04	14	MSAR vs SETAR			12	false

SKU	Horizon	Comparison	DM statistic	p-value	n	Significant
AG04	14	MSAR vs NAIVE	-0.584	0.2843	15	false
AJ00	1	MSAR vs ARIMA	-0.564	0.2886	28	false
AJ00	1	MSAR vs ETS	0.189	0.5741	28	false
AJ00	1	MSAR vs SETAR	-0.859	0.2011	18	false
AJ00	1	MSAR vs NAIVE	-1.303	0.1018	28	false
AJ00	7	MSAR vs ARIMA	0.896	0.8099	22	false
AJ00	7	MSAR vs ETS	0.618	0.7282	22	false
AJ00	7	MSAR vs SETAR	-1.004	0.1663	15	false
AJ00	7	MSAR vs NAIVE	0.206	0.5808	22	false
AJ00	14	MSAR vs ARIMA	-0.020	0.4923	15	false
AJ00	14	MSAR vs ETS	-0.086	0.4664	15	false
AJ00	14	MSAR vs SETAR	0.000	0.5000	14	false
AJ00	14	MSAR vs NAIVE	-0.238	0.4075	15	false
AG01	1	MSAR vs ARIMA	-1.623	0.0581	28	false
AG01	1	MSAR vs ETS	-1.675	0.0528	28	false
AG01	1	MSAR vs SETAR	0.249	0.5968	18	false
AG01	1	MSAR vs NAIVE	-1.306	0.1013	28	false
AG01	7	MSAR vs ARIMA	-0.978	0.1695	22	false
AG01	7	MSAR vs ETS	-1.321	0.1004	22	false
AG01	7	MSAR vs SETAR	-0.720	0.2423	14	false
AG01	7	MSAR vs NAIVE	-0.698	0.2465	22	false
AG01	14	MSAR vs ARIMA	-0.367	0.3595	15	false
AG01	14	MSAR vs ETS	-0.203	0.4209	15	false
AG01	14	MSAR vs SETAR			8	false
AG01	14	MSAR vs NAIVE	-0.385	0.3529	15	false

Detailed Model Confidence Set Results

Table XVII: Complete Model Confidence Set results

SKU	Horizon	Surviving models	Observations	Status
AA00	1	ARIMA, MSAR, NAIVE, SETAR	23	ok
AA00	7	ARIMA, MSAR, SETAR	18	ok
AA00	14	ARIMA, MSAR, SETAR	15	ok
AG00	1	ARIMA, ETS, MSAR, NAIVE, SETAR	15	ok
AG00	7	MSAR, SETAR	13	ok
AG00	14	insufficient data	9	insufficient data
AA03	1	ARIMA, ETS, MSAR, NAIVE, SETAR	20	ok
AA03	7	ARIMA, ETS, MSAR, SETAR	19	ok
AA03	14	ARIMA, ETS, MSAR, SETAR	12	ok
AA04	1	ARIMA, ETS, MSAR, NAIVE, SETAR	16	ok
AA04	7	MSAR, SETAR	12	ok
AA04	14	MSAR, SETAR	12	ok
AG03	1	ARIMA, ETS, MSAR, NAIVE, SETAR	14	ok
AG03	7	ARIMA, ETS, MSAR, NAIVE, SETAR	16	ok
AG03	14	MSAR, NAIVE, SETAR	13	ok
AG04	1	ARIMA, ETS, MSAR, NAIVE, SETAR	13	ok
AG04	7	MSAR	18	ok
AG04	14	MSAR, SETAR	12	ok
AJ00	1	ARIMA, ETS, MSAR, NAIVE, SETAR	18	ok
AJ00	7	ARIMA, ETS, MSAR, NAIVE, SETAR	15	ok
AJ00	14	ARIMA, ETS, MSAR, NAIVE, SETAR	14	ok
AG01	1	ARIMA, ETS, MSAR, NAIVE, SETAR	18	ok
AG01	7	MSAR, NAIVE, SETAR	14	ok
AG01	14	ETS, MSAR	15	ok

Low-Stock Threshold Robustness

Table XVIII: Monte Carlo robustness across alternative low-stock thresholds

SKU	Quantile	Threshold	h = 1	h = 7	h = 14
AA00	5th pct	4,448	0.654	0.218	0.112
AA00	10th pct	4,645	0.690	0.262	0.150
AA00	20th pct	4,977	0.756	0.363	0.228
AG00	5th pct	2,890	0.500	0.332	0.196
AG00	10th pct	4,307	0.925	0.581	0.470
AG00	20th pct	4,549	0.929	0.610	0.532
AA03	5th pct	2,329	0.050	0.144	0.165
AA03	10th pct	2,489	0.072	0.226	0.203
AA03	20th pct	2,755	0.798	0.390	0.276
AA04	5th pct	2,125	0.009	0.096	0.098
AA04	10th pct	2,648	0.901	0.390	0.258
AA04	20th pct	2,748	0.909	0.439	0.304
AG03	5th pct	1,889	0.028	0.075	0.075
AG03	10th pct	2,260	0.774	0.302	0.194
AG03	20th pct	2,345	0.788	0.340	0.228
AG04	5th pct	1,548	0.604	0.133	0.035
AG04	10th pct	2,196	0.670	0.280	0.208
AG04	20th pct	2,255	0.678	0.310	0.243
AJ00	5th pct	1,720	0.034	0.088	0.157
AJ00	10th pct	1,751	0.040	0.124	0.172
AJ00	20th pct	1,806	0.052	0.312	0.204
AG01	5th pct	896	0.015	0.044	0.050
AG01	10th pct	1,173	0.794	0.380	0.264
AG01	20th pct	1,217	0.806	0.419	0.310

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